

How Much History Do Deep Sequence Models Need for AIS Trajectory Prediction?

Final Project Report - Deep Learning Course 046211

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Abstract

This project studies vessel trajectory prediction from Automatic Identification System (AIS) data as a sequence learning problem. Given a history of position, speed, and direction sampled every 10 minutes, we predict the vessel latitude and longitude for the next 12 hours. The focus is not a maritime application by itself, but a central deep learning question: how much temporal context is useful for a sequential model, and does a more complex architecture exploit long history better?

We compare constant-speed and constant-course kinematic extrapolation, fixed-context AR LSTM models with history windows of 9, 12, 18, and 24 hours, a Flat LSTM that predicts the entire future trajectory directly, a Transformer, a short-window rolling model, and an Adaptive Multi-Scale RNN that learns soft weights over several history lengths. The main result is that neural models reduce the median final displacement error (FDE) from 102.5 km for the kinematic baseline to roughly 19-20 km. The best model is a 24-hour Flat LSTM, with median FDE of 18.80 km; the Transformer follows with 19.40 km. In the fixed AR LSTM sweep, 18 hours is best, while 24 hours does not improve the result. The adaptive model does not beat the simpler models: its gate almost always selects the 24-hour context and partly behaves as a geographic prior. Thus, long history is important, but useful adaptive history selection does not emerge automatically from simply adding a gate.

1 Introduction and Motivation

The Automatic Identification System (AIS) broadcasts vessel information such as geographic position, speed over ground, course over ground, heading, and time. These measurements form a time sequence, so vessel trajectory prediction is naturally a sequence modeling task. In this project we predict a full 12-hour trajectory, not only a nearby point. This creates a long-horizon setting in which a small early error may accumulate into a large final error.

From the perspective of the deep learning course, the task touches several core ideas. RNN/LSTM models compress the past into a hidden state; autoregressive prediction uses previous predictions to generate future ones; Transformers replace recurrent memory with self-attention; and regularization, loss design, teacher forcing, and curriculum learning affect training stability. Therefore, the question is not only which model performs best, but what the architectural and training choices teach us about learning long-range temporal dependencies.

The central research question is: how much history is useful for predicting a vessel trajectory 12 hours ahead? We test whether 24 hours is always better than 18 or 12, whether a Transformer exploits long context better than an LSTM, and whether an adaptive model can learn when to use short history and when to use long history. The main contributions are: a clean formulation of AIS forecasting as a long-sequence prediction problem; a comparison of architectures under the same data and metrics; a controlled sweep over history-window length; and an analysis of the learned gate weights in the adaptive model.

2 Data and Window Construction

The data is based on public NOAA AIS files from the United States, processed and merged into the USA Combined collection. Each record contains at least a geographic location, time, speed over ground (SOG), course over ground (COG), and sometimes heading. Since the original measurements are not uniformly spaced in time, trajectories were resampled to a fixed temporal resolution of $\Delta t = 10$ minutes.

Figure 1 illustrates why this task is not merely simple point regression: even when prediction begins at the same anchor point, a small deviation in speed or direction can become a large displacement after 12 hours. For this reason, the report focuses on trajectory metrics such as FDE and ADE rather than only instantaneous error.

Each training example is a continuous window. We store up to 24 hours of history, i.e. $T_h = 144$ steps, and predict 12 hours into the future, i.e. $T_f = 72$ steps. For fixed-context experiments we take a suffix of the full history window: 9 hours are 54 steps, 12 hours are 72 steps, 18 hours are 108 steps, and 24 hours are 144 steps. In all cases, the anchor point is the last observed position before prediction begins, so all models start from the same time point.

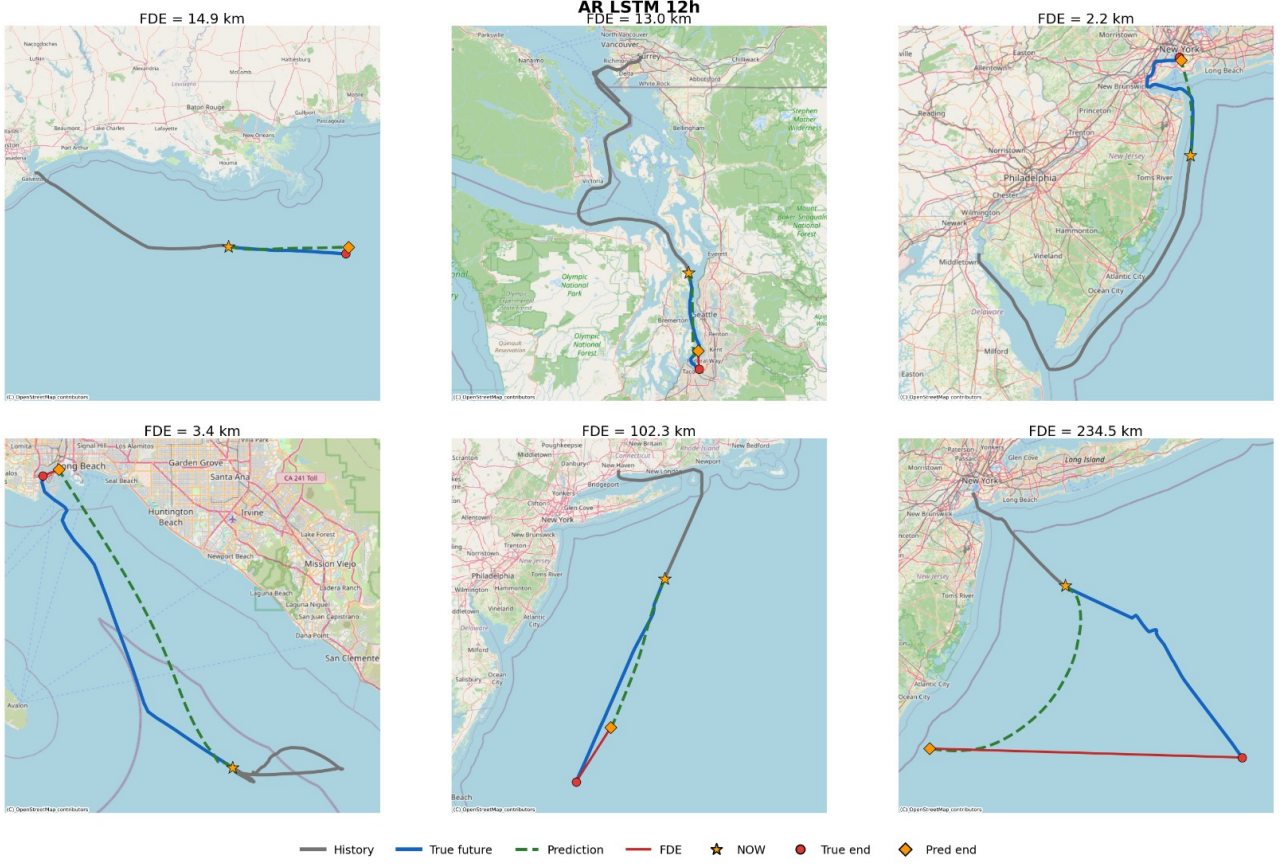


Figure 1: Example predictions from the test set. The gray curve is the vessel history, the blue curve is the true future trajectory, the green dashed curve is the model prediction, and the red segment marks the final displacement error.

The feature vector at each step includes position, speed, sine and cosine encodings of course, heading encoding, position differences, speed differences, course differences, and north/east velocity. Using sine and cosine for angles avoids an artificial discontinuity between 359° and 0° . Instead of predicting only absolute coordinates, the model predicts offsets from the anchor point:

$$y_t = P_t - P_0, \quad \hat{P}_t = P_0 + \hat{y}_t, \quad t = 0, \dots, 71.$$

This formulation reduces dependence on absolute location and allows the model to focus on the local motion pattern. Filtering was performed to focus on coastal and open-water trajectories. Nearly stationary windows were removed using thresholds on motion radius, total displacement, and mean speed. A smart-motion filter kept windows with meaningful movement and removed abnormal loops. Finally, inland rivers and canals were removed when more than half of the history points were not near open water. After filtering, 363,014 windows remained out of 535,967, i.e. 67.7% of the examples. The train/validation/test split was done by trajectory ID rather than by random rows, to avoid leakage between similar windows from the same voyage.

3 Prediction Target and Evaluation Metrics

The target is the vessel position over the next 12 hours at 10-minute resolution, so each example has $T_f = 72$ future steps. Let the anchor point, the last known position before prediction, be $P_0 = (\text{lat}_0, \text{lon}_0)$. Instead of predicting absolute position directly, the model predicts offsets from the anchor:

$$y_t = P_t - P_0, \quad \hat{P}_t = P_0 + \hat{y}_t, \quad t = 0, \dots, T_f - 1.$$

Here $P_t = (\text{lat}_t, \text{lon}_t)$ is the true future position and $\hat{P}_t$ is the predicted position. The notation lat denotes latitude and lon denotes longitude.

The main distance measure is the geodesic distance between prediction and ground truth, computed with the Haversine formula in kilometers. This formula measures the shortest distance on the Earth surface between two points given by latitude and longitude. If $P_1 = (\phi_1, \lambda_1)$ and $P_2 = (\phi_2, \lambda_2)$, where ϕ is latitude and λ is longitude in radians, then

$$\begin{aligned} \Delta\phi &= \phi_2 - \phi_1, & \Delta\lambda &= \lambda_2 - \lambda_1, \\ a &= \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cos(\phi_2) \sin^2\left(\frac{\Delta\lambda}{2}\right), \\ d_{\text{Hav}}(P_1, P_2) &= 2R \arcsin(\sqrt{a}), & R &\simeq 6371 \text{ km}. \end{aligned}$$

Final Displacement Error (FDE) measures the error at the last prediction step, i.e. after 12 hours:

$$\text{FDE} = d_{\text{Hav}}\left(\hat{P}_{T_f-1}, P_{T_f-1}\right).$$

Average Displacement Error (ADE) measures the mean error along the entire predicted trajectory:

$$\text{ADE} = \frac{1}{T_f} \sum_{t=0}^{T_f-1} d_{\text{Hav}}\left(\hat{P}_t, P_t\right).$$

The main ranking in this report uses median FDE, not only the mean, because long-horizon trajectory errors are heavy-tailed: a small number of abnormal voyages can strongly increase the average and give a less stable picture. We also report mean FDE and median ADE, because they help distinguish between a model that is accurate throughout the path and a model that is accurate mainly at the final point. In addition, examples were split into motion buckets, mainly straight versus maneuver. This checks whether the same history length is suitable for every trajectory type.

4 Models and Training Methods

4.1 Kinematic Baseline

The simple baseline continues the last observed SOG and COG for 12 hours. This is not a deep learning model, but it is important: if a neural network cannot beat such extrapolation, its complexity is not justified. In practice, the baseline is weak for a 12-hour horizon because vessels change speed, turn, enter and leave routes, and are affected by coastal geography.

4.2 Fixed-Context AR LSTM

The main model in the context-length experiment is an encoder-decoder LSTM. The encoder reads the history and produces a hidden state, and the decoder predicts position offsets step by step:

$$h_{\text{enc}} = \text{LSTM}_{\text{enc}}(x_{1:T_h}), \quad \hat{y}_t = f_{\text{dec}}(\hat{y}_{<t}, h_{\text{enc}}).$$

We trained the same architecture with four history lengths: 9, 12, 18, and 24 hours. In this experiment all other settings remain fixed, so the comparison isolates the effect of temporal context length.

4.3 Additional Models

- **Flat LSTM** reads 24 hours of history and predicts all 72 future points at once. Its advantage is that it avoids autoregressive error accumulation.
- **Transformer** uses self-attention over the long history, testing whether explicit attention over all steps is more useful than recurrent memory.
- **Sliding 3h x4** trains a short 3-hour predictor and rolls it four times to reach 12 hours. It updates the context at each rollout, but may accumulate errors.
- **Adaptive Multi-Scale RNN** encodes 9, 12, 18, and 24 hour windows and combines them using a soft gate:

$$\alpha = \text{softmax}(\text{MLP}([h_9; h_{12}; h_{18}; h_{24}])), \quad h_{\text{ctx}} = \sum_k \alpha_k h_k.$$

The purpose is to test whether the model can choose short or long context depending on the trajectory.

4.4 Loss Function and Training

It is important to distinguish evaluation metrics from the training objective. FDE and ADE are used to compare models after prediction, while the loss is the expression minimized during training. We use a combined objective: a Huber term on coordinate-offset error, a geographic distance term in kilometers, and a soft land penalty:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N w^{(i)} \left[0.5 L_{\text{Huber}}^{(i)} + 0.5 L_{\text{geo}}^{(i)} \right] + \lambda_{\text{land}} L_{\text{land}}.$$

The Huber term measures the error between predicted and true trajectory offsets. It is more stable than MSE on noisy AIS data: for small errors it behaves like a quadratic loss and gives smooth gradients, while for large errors it behaves more like a linear loss and prevents a few outliers from dominating training:

$$\text{Huber}_{\delta}(e) = \begin{cases} \frac{1}{2}e^2, & |e| \leq \delta, \\ \delta \left(|e| - \frac{1}{2}\delta \right), & |e| > \delta. \end{cases}$$

The geographic term L_{geo} gives coordinate errors a physical meaning in kilometers:

$$L_{\text{geo}}^{(i)} = \frac{1}{T} \sum_{t=0}^{T-1} d_{\text{geo}}(\hat{P}_t^{(i)}, P_t^{(i)}).$$

Here d_{geo} is a local kilometer-distance approximation that accounts for the latitude of the region. Its goal is not to replace FDE and ADE, but to help the training objective feel real distance rather than only numerical differences in degrees. In addition, L_{land} softly penalizes predictions that pass through land, encouraging geographically plausible maritime trajectories.

For autoregressive models we also use scheduled teacher forcing. Early in training, the decoder sometimes receives the true previous step; later, this probability is reduced until the model learns to use its own predictions. We also use horizon curriculum: training begins with a shorter prediction horizon and gradually grows to the full 12 hours. These two techniques reduce early training collapse and help the model handle error accumulation over long trajectories.

5 Experimental Design

The first experiment asks whether neural models learn beyond constant-speed and constant-course motion. Therefore, all models are compared against the kinematic baseline. The second experiment sweeps history length for AR LSTM: 9, 12, 18, and 24 hours. This is the cleanest deep learning experiment, because it directly studies temporal context while keeping the architecture fixed. The third experiment compares architectures: autoregressive prediction versus direct prediction, LSTM versus Transformer, and short rolling prediction versus direct long-horizon prediction. The fourth experiment analyzes the adaptive model: whether the context gate really chooses different windows, and what drives that choice.

Unless otherwise noted, all results refer to the filtered coastal data with land-penalty weight $\lambda_{\text{land}} = 0.1$. The main metric is median FDE after 12 hours.

6 Results

6.1 Overall Model Ranking

The overall ranking shows that the kinematic baseline obtains median FDE of 102.5 km, while the main neural models are in the range 18.8-22.4 km. Thus, deep models reduce final displacement error by roughly a factor of five. To compare the architectures directly, we rank all models by 12-hour median FDE on the same coastal test set. Figure 2 presents the central model comparison.

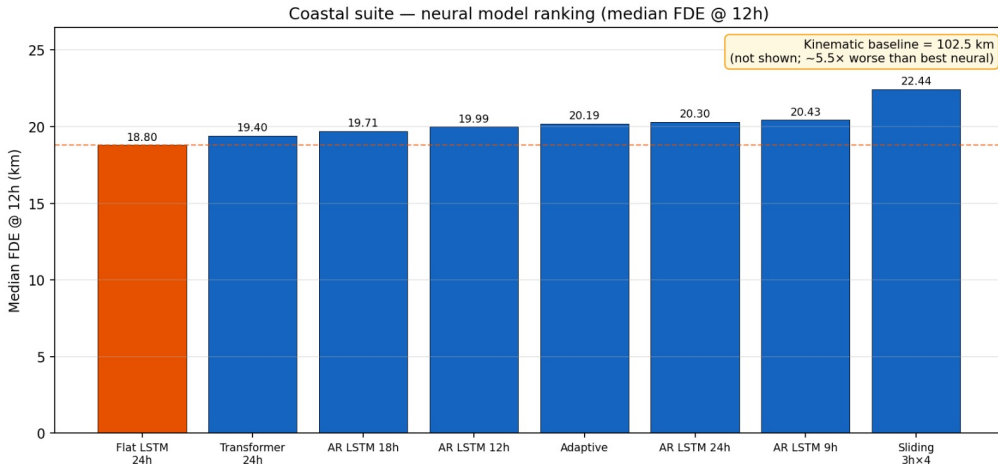


Figure 2: Model ranking by 12-hour median FDE on the coastal test set. The direct Flat LSTM obtains the best result, followed by the Transformer. All deep models are much better than the kinematic baseline, which is not shown on the full scale.

The large gap between the kinematic baseline and all neural models shows that the networks are not merely smoothing trajectories; they learn motion patterns that are absent from last-speed and last-course extrapolation. The first interesting result is that the relatively simple Flat LSTM is best. A possible explanation is that direct prediction of the full 12-hour trajectory avoids autoregressive error accumulation. In an autoregressive model, an early error becomes input to the next step and can grow along the horizon. The Transformer is second, showing that attention over long history is useful, but not necessarily better than the simpler Flat LSTM in this setting.

6.2 History-Length Sweep in AR LSTM

History length matters, but not monotonically. Moving from 9 to 12 hours improves median FDE; 18 hours gives the best result; and 24 hours slightly worsens it. Therefore, the answer to "is more history always better?" is no. After the overall comparison, we ran a controlled experiment on history length only: the same AR LSTM architecture, the same data, and the same training objective were used, while only the number of history hours changed.

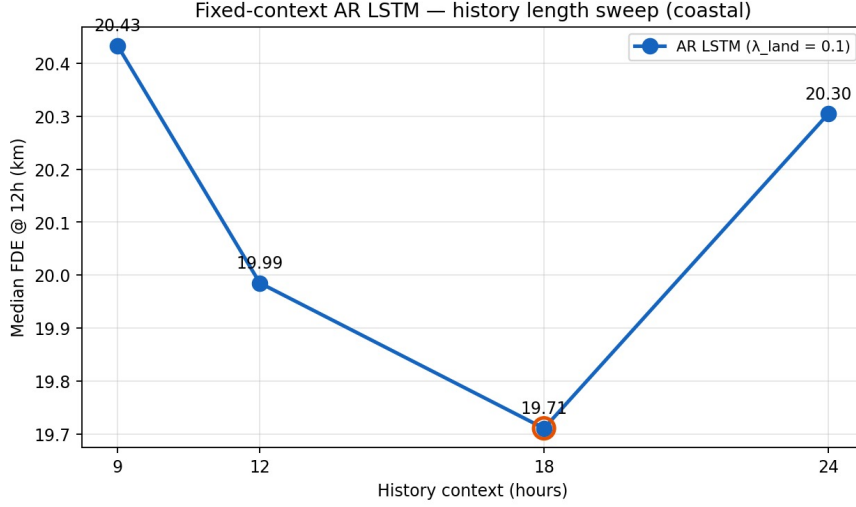


Figure 3: History-length sweep for the AR LSTM model. The error decreases from 9 to 18 hours, but increases again at 24 hours. More history is therefore not necessarily better; there is a balance between useful long-term information and old information that may add noise.

A possible interpretation is that the LSTM has a bottleneck: the whole history is compressed into a final hidden state. Very old information may belong to a motion regime that is no longer relevant, so it can add noise or make optimization harder. Eighteen hours may be long enough to capture route and speed information, but not long enough to overload the encoder with outdated data.

6.3 Analysis by Motion Type

Bucket analysis shows a more nuanced story than the overall ranking. For maneuvering trajectories, 9 hours of history is best; for straight trajectories, 18 hours is best. To test whether the 18-hour result comes uniformly from all trajectory types, we split the examples into two motion buckets: relatively straight trajectories and maneuvering trajectories. Figure 4 shows that the preferred history length depends on the type of motion.

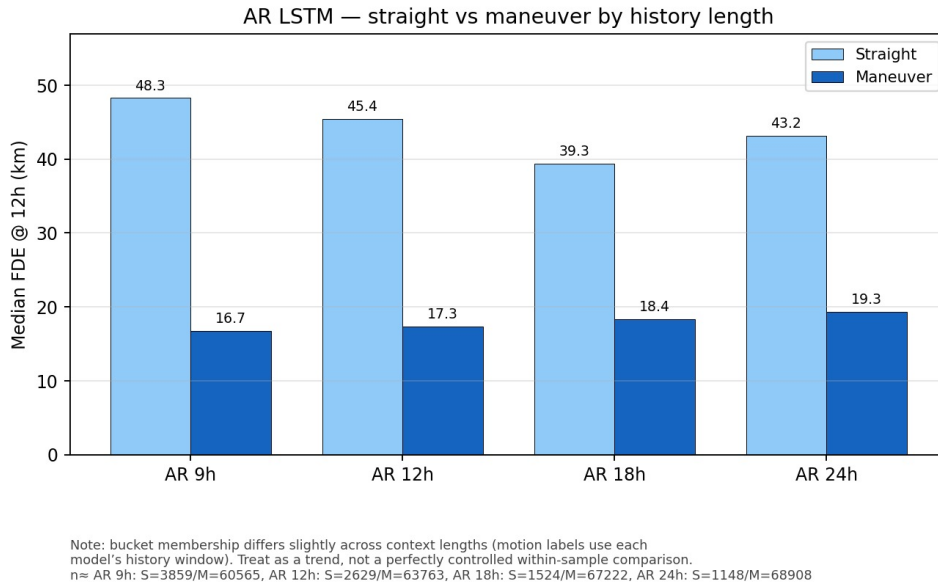


Figure 4: Median FDE for straight and maneuvering trajectories as a function of history length in the AR LSTM model. Straight trajectories improve clearly up to 18 hours, whereas for maneuvering trajectories shorter context remains competitive. The optimal history length is therefore not uniform across motion types.

Thus, the overall 18-hour optimum hides an internal difference: straight trajectories benefit from longer context, while for maneuvering trajectories old information may be less relevant. This result is important because it shows that there is no universal history length. A maneuvering vessel is strongly affected by its most recent state; a recent change of course or speed matters more than older information. In contrast, for a straight trajectory, longer history may help

identify a stable maritime route. This observation motivated the adaptive model.

6.4 Adaptive Model Result

The adaptive model was built to test whether history length can be learned from the data instead of fixed in advance for all vessels. In practice, we use one shared LSTM encoder for four nested windows of 9, 12, 18, and 24 hours. Each window produces a hidden representation h_H . The four representations are concatenated into $z = [h_9; h_{12}; h_{18}; h_{24}]$ and passed through a small MLP gate: a linear layer, ReLU, dropout, and a final linear layer that outputs four scores s_H . These scores are not the final weights; they pass through a softmax, producing positive weights α_H that sum to one:

$$\alpha_H = \frac{e^{s_H}}{\sum_{K \in \{9, 12, 18, 24\}} e^{s_K}}, \quad \sum_H \alpha_H = 1,$$

$$h_{\text{ctx}} = \sum_{H \in \{9, 12, 18, 24\}} \alpha_H h_H.$$

Conceptually, this is similar to an attention mechanism: there are several possible representations of the same history, and the model learns how much to look at each one. The difference is that this is not self-attention over all time steps, but attention-like pooling over four temporal scales. The gate does not replace the trajectory predictor; it only chooses the mixture of history representations that is fed into the autoregressive decoder.

The hypothesis was that different trajectories would choose different contexts: maneuvers or speed changes would prefer shorter history, while stable voyages along a maritime route would prefer longer history. To test whether the adaptive model really learns such a rule, we examine the gate weights on the coastal test set. If the model had learned a strong adaptive rule, we would expect a meaningful spread among 9, 12, 18, and 24 hours.

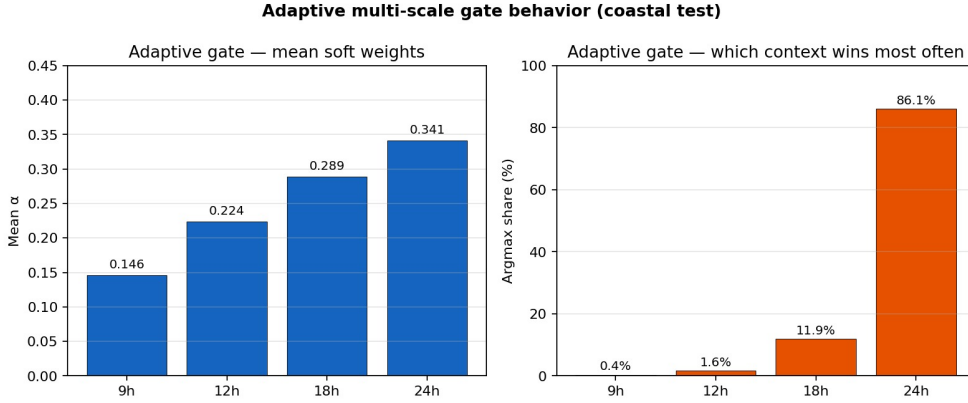


Figure 5: Behavior of the adaptive gate on the coastal test set. Left: mean α weights for each history window. Right: which window receives the largest weight in each example. Although the soft weights mix several windows, the 24-hour window wins in about 86% of the examples. The gate therefore does not learn a sharp transition between short and long context by motion type.

In practice, Figure 5 shows that the gate is strongly biased toward the 24-hour window. Thus, the adaptive result does not indicate true flexible switching between short and long history, but rather a general preference for long context. Feature analysis supports this interpretation: Spearman correlations between motion features and α were relatively weak. Distance and speed over 24 hours slightly increased the weights on 12-18 hours; speed changes slightly increased the 9-hour weight; and maneuvers slightly decreased the 18-hour weight. In contrast, a random forest model trained to explain the gate weights found that the most important features were the latitude and longitude of the anchor point. The conclusion is that the gate largely learned a geographic prior. It used the geographic region as a convenient cue for choosing a context mixture, more than a clean motion rule such as "maneuver needs short context" and "straight route needs long context." For an adaptive model to win, it may need a stronger inductive bias, a time-dependent gate, or regularization that forces real specialization between history windows.

7 Discussion

The first conclusion is that neural models do learn dynamics that are absent from the kinematic baseline. Reducing median FDE from 102.5 km to roughly 19 km shows that 12-hour prediction cannot be solved well by simply continuing the last speed and direction. The model must learn traffic routes, speed changes, turns, and geographic constraints.

The second conclusion is that more history is not necessarily better. In the AR LSTM sweep, 18 hours is better than 24 hours. This is important for the course because it demonstrates a limitation of recurrent memory: even if the network receives more time steps, there is no guarantee that it will use them beneficially. Old information may require attention, filtering, or dedicated regularization.

In addition to final-error ranking, we also examine how error grows over the 12-hour horizon. This is useful because two models may have similar final FDE but behave differently during the forecast. Figure 6 shows that errors generally increase as the prediction moves farther away from the anchor point.

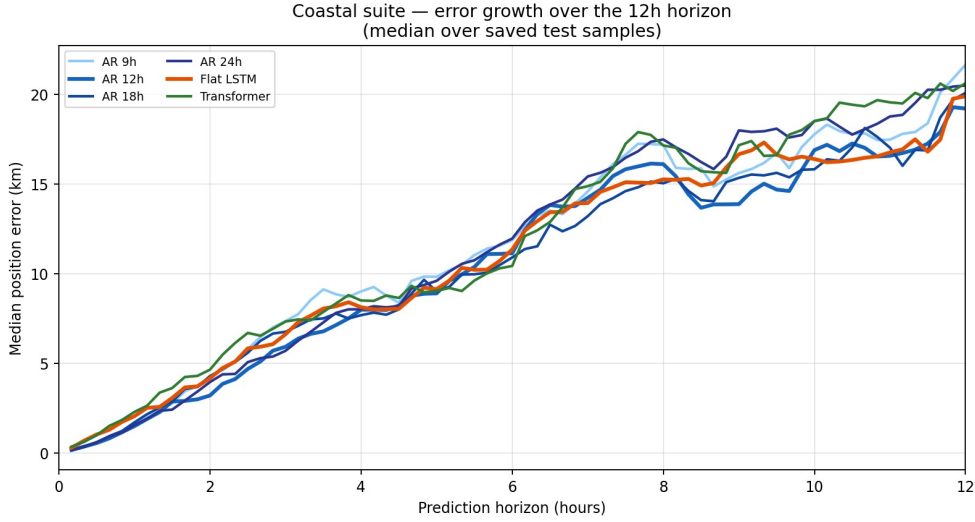


Figure 6: Growth of position error along the 12-hour forecast horizon. The error increases gradually as the prediction moves farther from the anchor point, so long-horizon forecasting is substantially harder than predicting only nearby steps.

The third conclusion is that direct prediction can be preferable to autoregressive prediction. Flat LSTM obtains the best result, probably because it avoids accumulated errors over 72 prediction steps. On the other hand, autoregressive modeling is a natural way to generate a sequence, so it remains important for comparison and analysis.

The fourth conclusion is that adaptivity is not a magic solution. The adaptive gate almost always selects 24 hours and does not learn a strong rule that switches between short and long context based on maneuvering. To make such a model successful, it may be necessary to use a gate per future step, a penalty that encourages diverse choices, more separate experts, or explicit motion features in the gate rather than relying only on encoder states.

8 Limitations and Future Work

The project has several limitations. First, the land mask is relatively coarse, so the land penalty does not replace a full geographic model of shipping lanes and ports. Second, the main results are all on USA Combined after coastal filtering; in other regions or inland routes, the model ranking may change. Third, we did not run a full hyperparameter sweep for every model, so the comparison is between well-trained models but not a global optimization.

Natural future work includes an ablation of the land-penalty weight, testing more history lengths for the Transformer, and adding explicit geographic features such as distance from shore, distance to port, or local AIS density. For the adaptive model, it would be useful to test a gate that also depends on future prediction step, or a gate that directly receives motion features such as straightness, COG variance, and SOG variance. Such work would test whether adaptivity can be turned from a general geographic mechanism into a mechanism that truly understands trajectory dynamics.

9 Ethics Statement

Student names: Amitai Gal and [Partner Name].

Project title: How Much History Do Deep Sequence Models Need for AIS Trajectory Prediction?

Project description: This project develops and compares deep learning models for predicting vessel trajectories from public AIS data, for academic research on sequence learning. The models are not intended for operational use in navigation, safety, or enforcement, and their predictions are approximate.

Stakeholders. Three central stakeholder groups are coastal and maritime-safety authorities, shipping companies and ports, and vessel crews or operators. Authorities may use forecasts to understand traffic load and collision risk; companies and ports may use them for planning arrivals and departures; and crews may be affected if incorrect predictions are used for decision making. There is also a possible privacy and surveillance impact, because movement data may reveal patterns of activity.

Explanation and responsibility. Authorities should be told that the model is a statistical tool and not a certified safety system, so it should not be used alone for critical decisions. Shipping companies and ports should receive clear error metrics, limits of use, and examples of failures. Vessel crews should be told that the model does not replace human navigation, radar, warning systems, or safety procedures. The responsibility for these explanations belongs to both the system developers and the organization that decides to deploy it; in this project, our responsibility is to report limitations, biases, and errors honestly.

Reflection. In our reflection, the main risk is not only prediction error but also misuse of the model. Even if the model has good average error, a single failure case may be significant if used for safety, enforcement, or surveillance.

Therefore, an ethical explanation should present not only average performance, but also uncertainty, failure cases, geographic limitations, and a clear restriction against using the model as a replacement for certified navigation and control systems.

10 Conclusion

The project shows that 12-hour AIS trajectory prediction is a challenging sequence learning problem, but neural models improve greatly over simple kinematic extrapolation. The best result is obtained by a 24-hour Flat LSTM, while the Transformer is close behind. In the AR LSTM sweep, 18 hours is better than 24, so longer history is not always useful. The adaptive model provides an important lesson: although it is more flexible, it tends to choose 24 hours almost always and is strongly affected by geography. The main conclusion is that temporal context matters, but learning how much history to use requires more than a simple gating mechanism.

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